

Thermal Transport in Cylindrical Channels

Non-Uniform Wall Boundary Conditions

and Large Aspect Ratio Analysis

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Introduction

These notes develop a perturbative analysis of how thermal non-uniformity affects the flow profile in laminar cylindrical channels, motivated by transformer cooling applications.

The goal is to address a gap in the electrification literature. Analyses of charge transport in transformer oil typically assume isothermal flow with constant physical properties (viscosity, thermal and electrical conductivity, density). We relax this assumption and study the leading-order effects of temperature-dependent properties on both flow and later transport.

We acknowledge the limitations: real transformer temperature gradients are often too large for perturbation theory to be quantitatively accurate. This work is not meant to replace CFD simulation, but rather to provide an analytical framework for understanding first-order coupling mechanisms and guiding numerical predictions.

To achieve this goal, we started by solving the isothermal flow profile. This result is then used in the energy equation to derive the temperature distribution in the fluid under non homogeneous boundary conditions. These results are coupled back to the momentum equations through the fluid properties dependence on temperature allowing for a computation of the first order correction to the velocity fluid.

Notation. Starred quantities (T^* , r^* , z^* , etc.) denote dimensionless variables. The asterisk is sometimes suppressed for brevity. The Laplacian operator in cylindrical coordinates appears as ∇^2 (full 3D) or $\nabla_{\perp}^2$ (radial + angular only).

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I Isothermal Velocity Field

I.1 Physical Setup & Symmetry Considerations

Consider a long cylindrical cooling channel of radius R and length L , filled with a viscous fluid (e.g., transformer oil). The fluid enters at a uniform inlet temperature and flows under a pressure gradient Δp . Table I summarizes the different physical properties of the fluid and the symbols used to represent them.

Physical property	symbol
Density	ρ
viscosity	μ
Thermal Capacity per unit mass at constant pressure	$\hat{C}_p$
Thermal conductivity	K
Thermal diffusivity	α

Table I: Summary of the fluid physical properties and symbols

Assuming that the length of the cooling channel L is much larger than the radius R , we expect that the velocity field to be predominantly axial along the z direction with only radial dependence. Hence for the isothermal case we need only consider $v_z(r)$.

I.2 Momentum transport & Velocity profile

We first start by defining the relevant mathematical objects for the derivation of the momentum transport equations. There are three different mechanism responsible for momentum transport for a flowing system

- First, transport through the bulk movement of the fluid. This is represented formally through *The Convective Momentum Transport tensor* $\phi_{ij} = \rho v_i v_j$.
- Second, momentum transport through molecular motion. This is taken into account through *The Shear Stress tensor*, τ_{ij} defined by

$$\tau_{ij} = -\mu \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) + \left(\frac{2}{3}\mu - \kappa \right) \nabla \cdot \mathbf{v} \delta_{ij}. \quad (1)$$

κ is the bulk viscosity of the fluid. For this analysis, we will only consider incompressible fluids verifying the equation $\nabla \cdot \mathbf{v} = 0$. Hence the second term in the previous expression vanishes.

- Finally, momentum can be transported through pressure gradients.

To take into account all three forms of momentum transport we define the *Combined Momentum Tensor*, Φ_{ij}

$$\Phi_{ij} = \rho v_i v_j + \tau_{ij} - P \delta_{ij}. \quad (2)$$

Physical Remark. The tensor element Φ_{ij} can be interpreted as the transport rate of the j -th component of momentum across a surface orthogonal to the i direction.

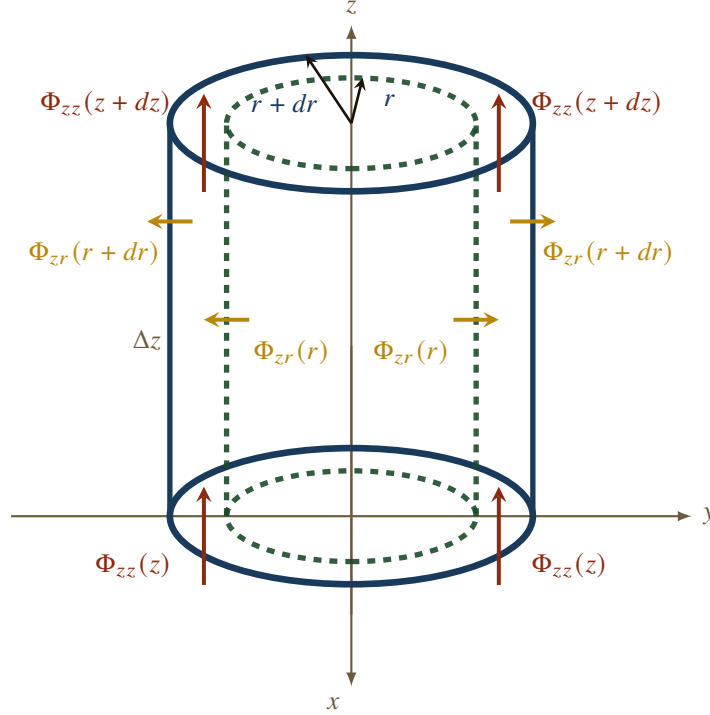


Figure 1: Cylindrical shell element showing the annular region between radius r (inner, dashed) and radius R (outer, solid) with axial extent Δz . The combined momentum tensor flux is shown.

The shell balance method consists of equating the rate of momentum change for a volume shell to the flux of the tensor through its boundaries. Given that we consider a steady state flow, we thus impose a null momentum balance through the shell boundaries.

Derivation 1.1 (Momentum Transport Equation). Consider the volume shell represented in figure 1. Given the symmetries of the system and the postulated velocity field form the only non vanishing terms of Φ are Φ_{zz} and ϕ_{zr} . The momentum balance for the z component is given by

$$2\pi r dr (\Phi_{zz}(z) - \Phi_{zz}(z + dz)) + (2\pi r dz \Phi_{rz}(r) - 2\pi(r + dr) dz \Phi_{rz}(r + dr)) = 0 \quad (3)$$

Dividing by the volume element $dv = 2\pi r dr dz$ and taking the limit $dr \rightarrow 0$, $dz \rightarrow 0$ leads to the following differential equation

$$-\frac{\partial \Phi_{zz}}{\partial z} - \frac{\partial \Phi_{rz}}{\partial r} - \frac{1}{r} \Phi_{rz} = 0. \quad (4)$$

Expressing the dependence of Φ explicitly on the velocity field,

$$\Phi_{zz} = \rho v_z(r)^2 - P(z), \quad (5a)$$

$$\Phi_{rz} = \tau_{rz} = -\mu \partial_r v_z, \quad (5b)$$

we get the momentum transport equation verified by the velocity field,

$$\boxed{+\partial_z P = -\mu \frac{1}{r} \partial_r (r \partial_r v_z)} \quad (6)$$

Integrating this equation leads to

$$v_z(r) = -\frac{\Delta P}{4\mu} r^2 + c \ln(r) + c'. \quad (7)$$

c and c' are integration constant. Their value is determined through the no slip boundary condition at $r = R$, $v_z(R) = 0$, and requiring the centerline velocity $v_z(0)$ is finite. Hence we get

$$\boxed{v_z(r) = v_0 \left(1 - \left(\frac{r}{R}\right)^2\right)}, \quad (8)$$

with $v_0 = \frac{\Delta P}{4\mu L} R^2$.

Physical Remark (remarks on Poiseuille Velocity Profile). Equation 8, is the standard Poiseuille flow result. The velocity field is parabolic with a maximum velocity at the center of the channel. The shear stress is maximum at the boundary, $r = R$, and equal to $\tau_{wall} = \frac{2\mu v_0}{R}$.

Regarding the derivation of the equation, one should note that, in general, the continuity equation ($\nabla \cdot \mathbf{v} = 0$ for incompressible fluids) is also needed. The shell balance techniques for momentum transport provides three equations, one for each on the momentum components. The total number of unknowns on the other hand is four, the three velocity components and pressure. In the derivation of the velocity field the pressure gradient along the z -direction was not specified nor solved for, this is because we assume an experimental setup where the pressure gradient is controlled.

2 Energy Balance & Temperature Profile

With the isothermal steady state velocity at hand, we now consider the case where the boundary of the channel, $r = R$, has a non-uniform temperature distribution both in the azimuthal and the longitudinal direction, $T(R, \theta, z) = T_{BC}(\theta, z)$. The non-uniform boundary condition break the axial symmetry.

2.1 Energy Transport Mechanisms & the First Law of thermodynamics

In this analysis we neglect temperature transport through radiation and focus mainly the the diffuse and convective mechanisms. Thus, let $\mathbf{q}$ be the *molecular diffusion energy flux*, it represents the energy

flux due to the thermal movement of molecules. It is related to the temperature field through Fourier's law, $\mathbf{q} = -K \nabla T$, with K the thermal conductivity of the fluid.

To derive the convective energy transport, consider a fluid volume dV moving with a velocity $\mathbf{v}$. The energy contained in this volume is given by

$$E = \left(\frac{1}{2} \rho v^2 + \rho \hat{U} \right) dV, \quad (9)$$

where $\hat{U}$ is the internal energy per unit volume of the fluid. the energy flux across a surface element $d\mathbf{s}$ orthogonal to the x direction is then

$$Q = \left(\frac{1}{2} \rho v^2 + \rho \hat{U} \right) v_x dV ds, \quad (10a)$$

$$= \mathbf{c} \cdot d\mathbf{s} \quad (10b)$$

Where $\mathbf{c}$ is the *convective energy flux*, $\mathbf{c} = \left(\frac{1}{2} \rho v^2 + \rho \hat{U} \right) \mathbf{v}$.

Finally, the derivation of the energy equation requires the application of the first law of thermodynamics for an open system. This is achieved by requiring that the total energy variation for a volume element is equal to the heat intake and to the work of the molecular forces. Consider a cube of volume dV and let ds be one of its sides at a constant x value, the force applied at this surface is $\Pi_x ds$ with $\Pi_x)_i = \tau_{xi} + P\delta_{xi}$. Where τ_x represents the shear forces applied at the surface and P the pressure. Thus the rate of the work applied at the surface is given by $\frac{dw}{dt} = \Pi_x \cdot \mathbf{v} ds$. Hence for a general surface give by the normal $d\mathbf{s}$. The work rate is $(\Pi \cdot \mathbf{v}) \cdot d\mathbf{s}$.

Remark. The operation $\cdot$ here is used in generalized form as a dot product between tensors of different ranks. Thus in the expression of the work rate it take a second rank tensor Π and a first rank tensor $\mathbf{v}$ and produces a first rank tensor.

In a similar manner to the *combined momentum transport tensor* Φ , we define the *combined energy flux vector* $\mathbf{e}$ by

$$\mathbf{e} = \left(\frac{1}{2} \rho v^2 + \rho \hat{U} \right) \mathbf{v} + \Pi \cdot \mathbf{v} + \mathbf{q}. \quad (11)$$

The flux of this vector field along a closed surface represents to the total energy variation of the surface. We shall express $\mathbf{e}$ in a more useful form using the enthalpy $\hat{H} = \hat{U} + P \hat{V}$, where $\hat{V} = \frac{1}{\rho}$ is the volume per unit mass of the fluid. Expanding Π in pressure and shear contribution, we get

$$\mathbf{e} = \left(\frac{1}{2} \rho v^2 + \rho \left(\hat{U} + P \hat{V} \right) \right) \mathbf{v} + \tau \cdot \mathbf{v} + \mathbf{q}, \quad (12a)$$

$$= \left(\frac{1}{2} \rho v^2 + \rho \hat{H} \right) \mathbf{v} + \tau \cdot \mathbf{v} + \mathbf{q} \quad (12b)$$

Derivation 2.1 (Temperature transport equation). Given the dependence of the boundary conditions on the azimuthal angle θ , we consider the shell shown in figure 2. The energy balance

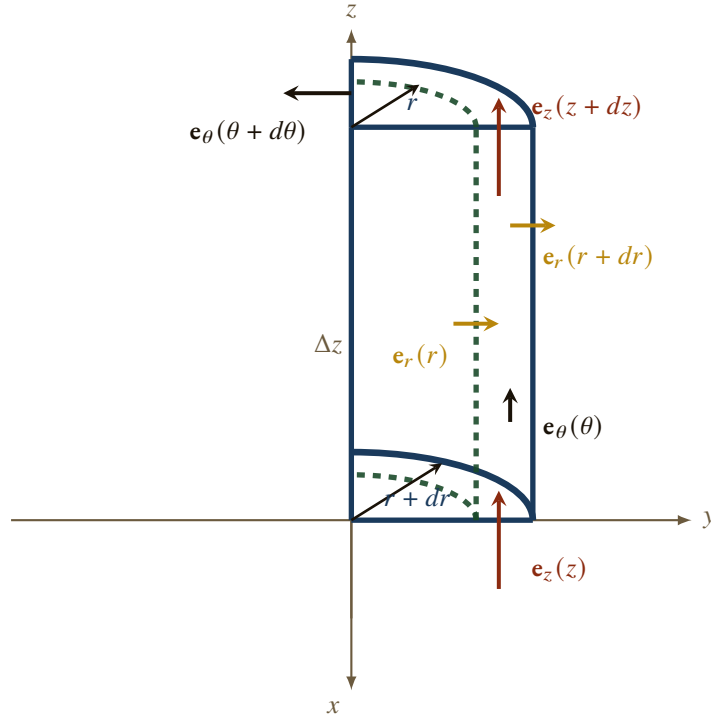


Figure 2: Combined energy flux through a cylindrical volume element.

equation is given by

$$r \, dr \, d\theta \, (e_z(z) - e_z(z + dz)) + r \, dz \, d\theta \, (e_r(r) - e_r(r + dr)) - dr \, d\theta \, dz \, e_r(r + dr) \quad (13a)$$

$$+ dr \, dz \, (e_\theta(\theta) - e_\theta(\theta + d\theta)) = 0. \quad (13b)$$

Dividing by the volume element $dV = r \, dr \, d\theta \, dz$ leads to

$$-\partial_z e_z - \frac{1}{r} \partial_r (r e_r) - \frac{1}{r} \partial_\theta e_\theta = 0 \quad (14)$$

Using the velocity profile 8, we can combine each derivative appearing in the energy balance equation.

- Given $e_z = \left(\frac{1}{2} \rho \mathbf{v}^2 + \rho \hat{H} \right) v_z + q_z$, we get

$$\partial_z e_z = \rho \partial_z \hat{H} + \partial_z q_z. \quad (15)$$

The dependence of the enthalpy on coordinates is implicit through its dependence on temperature and pressure. Thus,

$$\partial_z \hat{H} = \partial_z T \, \partial_T \hat{H} + \partial_z P \, \partial_P \hat{H}. \quad (16)$$

Using the following thermodynamics identities,

$$\partial_T \hat{H} = \hat{c}_p \quad , \quad \partial_P \hat{H} = \hat{V} + \rho \partial_P \hat{V}, \quad (17)$$

and assuming a constant fluid density, we get

$$\partial_z \hat{H} = \hat{c}_p \partial_z T + \partial_z P. \quad (18)$$

Fourier's law indicates that $q_z = -K \partial_z T$. Putting everything together we get

$$\partial_z e_z = (\rho \hat{c}_p \partial_z T + \partial_z P) v_z - K \partial_z^2 T \quad (19)$$

- For the radial component of the or the combined energy flux vector, we have

$$e_r = \tau_{rz} v_z + q_r \quad (20)$$

Thus

$$\frac{1}{r} \partial_r (r e_r) = -\frac{\mu}{2} \frac{1}{r} \partial_r (r \partial_r v_z^2) - \frac{K}{r} \partial_r (r \partial_r T) \quad (21)$$

- Finally, $e_\theta = q_\theta$, hence

$$\frac{1}{r} \partial_\theta e_\theta = -\frac{K}{r^2} \partial_\theta^2 T \quad (22)$$

The energy balance equation takes the form

$$K \nabla^2 T = \rho \hat{c}_p v_z \partial_z T + v_z \partial_z P - \frac{\mu}{2} \frac{1}{r} \partial_r (r \partial_r v_z^2). \quad (23)$$

This equation can be further simplified using the momentum transport equation 6.

$$v_z \partial_z P - \frac{\mu}{2} \frac{1}{r} \partial_r (r \partial_r v_z^2) = v_z \underbrace{\left[\partial_z P - \mu \frac{1}{r} \partial_r (r \partial_r^2 v_z) \right]}_{=0} - \mu (\partial_r v_z)^2 \quad (24)$$

Finally the heat equation takes the following form

$$\boxed{\rho \hat{c}_p v_z \partial_z T = K \nabla^2 T + \mu (\partial_r v_z)^2} \quad (25)$$

2.2 Dimensionless equation & Reference Scales

Before solving the heat transport equation 25, we first express it in a dimensionless form. To this end define the dimensionless parameters 2. Here we have defined the kinetic temperature $T_k = \frac{v_o^2}{2\hat{c}_p}$. In this new parameters the temperature transport equation takes the form

$$v^* \partial_{z^*} T^* = \frac{\alpha}{Lv_o} \left[\partial_{z^*}^2 T^* + \left(\frac{L}{R} \right)^2 \frac{1}{r^*} \partial_{r^*} (r^* \partial_{r^*} T^*) + \left(\frac{L}{R} \right)^2 \frac{1}{r^{*2}} \partial_{\theta^*}^2 T^* \right] + 8 \frac{\mu}{\rho Lv_o} \left(\frac{L}{R} \right)^2 r^{*2} \quad (26)$$

Variable	Expression
r^*	$\frac{r}{R}$
v^*	$\frac{v_z}{v_0}$
z^*	$\frac{z}{L}$
T^*	$\frac{T}{T_k}$

Table 2: Dimensionless parameter definition

The parameter $\alpha = \frac{K}{\rho \hat{c}_p}$ is the temperature diffusivity of the fluid. Finally using the definition of the Péclet, Reynolds, Prandtl numbers and the aspect ratio,

$$\text{Pe} = \frac{Lv_0}{\alpha}, \quad \text{Re} = \frac{\rho R v_0}{\mu}, \quad \epsilon = \frac{R}{L}, \quad \text{Pr} = \epsilon \frac{\text{Pe}}{\text{Re}} \quad (27)$$

we get the following equation

$$v^* \epsilon^2 \partial_{z^*} T^* = \frac{1}{\text{Pe}} \left[\epsilon^2 \partial_{z^*}^2 T^* + \frac{1}{r^*} \partial_{r^*} (r^* \partial_{r^*} T^*) + \frac{1}{r^{*2}} \partial_{\theta}^2 T^* \right] + \frac{8}{\text{Re}} \epsilon r^{*2}. \quad (28)$$

2.3 Small Aspect Ratio limit & Temperature Profile

Equation 28 indicates a clear scale separation between the azimuthal and radial variations of the temperature field and the longitudinal ones. Thus considering the case of small aspect ratio $\epsilon \rightarrow 0$ we can ignore the longitudinal variation of the temperature the equation takes then the simplified form.

$$\frac{1}{r^*} \partial_{r^*} (r^* \partial_{r^*} T^*) + \frac{1}{r^{*2}} \partial_{\theta}^2 T^* = -8 \text{Pr} r^{*2}. \quad (29)$$

Derivation 2.2 (Solving the Temperature Transport Equation). To solve the previous equation we first consider the homogeneous equation without the shear source

$$\frac{1}{r^*} \partial_{r^*} (r^* \partial_{r^*} T^*) + \frac{1}{r^{*2}} \partial_{\theta}^2 T^* = 0. \quad (30)$$

Separating the variables,

$$T^*(r^*, z^*, \theta) = f(r^*, z^*) g(\theta), \quad (31)$$

equation ?? separates into two equations

$$\frac{d^2 g}{d\theta^2} + \omega^2 g(\theta) = 0 \quad (32a)$$

$$\frac{1}{r^*} \partial_{r^*} (r \partial_{r^*} f) = \omega^2 f. \quad (32b)$$

The first equation leads to $g(\theta) = e^{i\omega\theta}$ then imposition the periodicity condition $g(\theta) = g(\theta + 2\pi)$ leads to $\omega = m \in \mathbb{Z}$. The second equation is solved by expressing f as power series $f(r^*, z^*) = \sum_{n \in \mathbb{Z}} a_n(z^*) (r^*)^n$, the second equation then imposes that $n^2 = m^2$. However the

demand that the temperature at $r^* = 0$ stay finite leads to $n = |m|$. Thus, a general solution for the homogeneous equation is

$$T^*(r^*, z^*, \theta) = \sum_{m \in \mathbb{Z}} a_m(z) r^{*|m|} e^{im\theta}. \quad (33)$$

To obtain a solution of the temperature transport equation we need to find a particular solution to 29. Thus, let $T_p(r^*)$ be a solution. We get

$$T_p(r^*) = -\frac{\text{Pr}}{2} r^{*4} \quad (34)$$

Hence the temperature field is

$$T^*(r^*, z^*, \theta) = -\frac{\text{Pr}}{2} r^{*4} + \sum_{m \in \mathbb{Z}} a_m(z) r^{*|m|} e^{im\theta}. \quad (35)$$

The coefficients $a_m(z)$ are determined through the boundary conditions by

$$a_m(z) = \begin{cases} a_0(z) = \frac{1}{2\pi} \int_0^{2\pi} T_{BC}(\theta, z) d\theta + \frac{\text{Pr}}{2} \\ a_m(z) = \frac{1}{2\pi} \int_0^{2\pi} T_{BC}(\theta, z) e^{-im\theta} d\theta, m \neq 0 \end{cases} \quad (36)$$

3 Test Cases and Validation

3.1 Case 1: Uniform Wall Temperature

Suppose the wall is isothermal: $T_w(\theta) = T_o$ (constant).

Then:

$$a_m = \frac{1}{2\pi} \int_0^{2\pi} (T_o + \frac{\text{Pr}}{2}) e^{-im\theta} d\theta = \begin{cases} (T_o + \frac{\text{Pr}}{2}) & m = 0 \\ 0 & m \neq 0 \end{cases} \quad (37)$$

The solution becomes:

$$T^* = \frac{\text{Pr}}{2} (1 - r^{*4}) + T_o. \quad (38)$$

This is purely antisymmetric and matches the classical result for Poiseuille flow with uniform wall temperature.

3.2 Case 2: One-Sided Cooling (Non-Uniform)

Suppose cooling is stronger on one side:

$$T_w(\theta) = T_o + \Delta T \cos(\theta). \quad (39)$$

Then:

$$a_m = \frac{1}{2\pi} \int_0^{2\pi} [\Delta T \cos \theta] e^{-im\theta} d\theta. \quad (40)$$

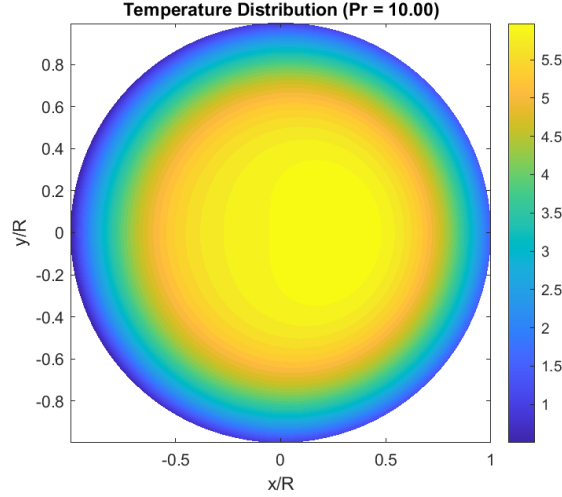


Figure 3: One-Sided Cooling Temperature Profile at $Pr = 10$, $T_o = 1$, $\Delta T = 0.5$,

By the orthogonality of Fourier modes:

$$a_m = \begin{cases} (T_o + \frac{Pr}{2}) & m = 0 \\ \Delta T / 2 & m = \pm 1 \\ 0 & |m| \geq 2 \end{cases} \quad (41)$$

The solution is:

$$T^* = \frac{Pr}{2} (1 - r^{*4}) + T_o + \frac{\Delta T}{2} (r^* e^{i\theta} + r^* e^{-i\theta}) = \frac{Pr}{2} (1 - r^{*4}) + T_o + \Delta T r^* \cos \theta. \quad (42)$$

Physical interpretation:

- The $r^{*4}/4$ term is the dissipation baseline.
- The constant $(T_o + \frac{Pr}{2})$ is the uniform cooling response.
- The $r^* \cos \theta$ term shows the asymmetry penetrates radially with amplitude proportional to r^* — it decays toward the axis.
- Higher harmonics ($m \geq 2$) are automatically zero, reflecting that the BC only has one-sided structure.

Figure 3 shows the dimensionless temperature profile for one sided cooling at $Pr = 10$.

3.3 Case 3: Non uniform Boundary Conditions with 3 different modes

To investigate more relativistic case we consider aa boundary conditions with 3 different modes,

$$T_{BC}^* = T_o + \Delta T \cos(\theta) + A_2 \cos(2\theta) + A_3 \cos(3\theta) \quad (43)$$

Figure 4 Shows the temperature distribution under boundary conditions given by equation 43. Com-

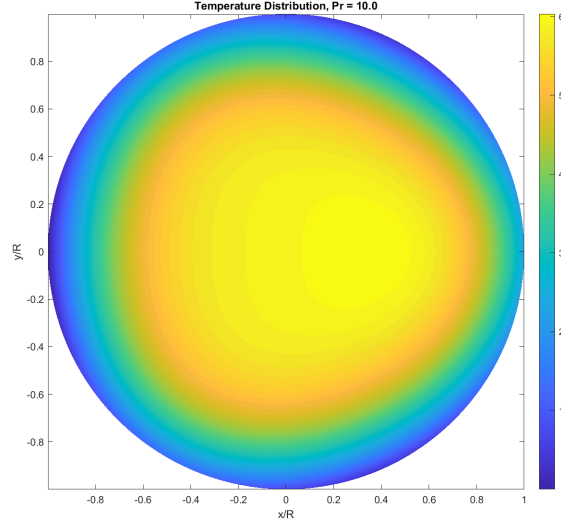


Figure 4: Temperature profile for multi-mode boundary condition. $Pr = 10$, $T_o = 1$, $A_1 = 0.5$, $A_2 = 0.2$, $A_3 = 0.7$

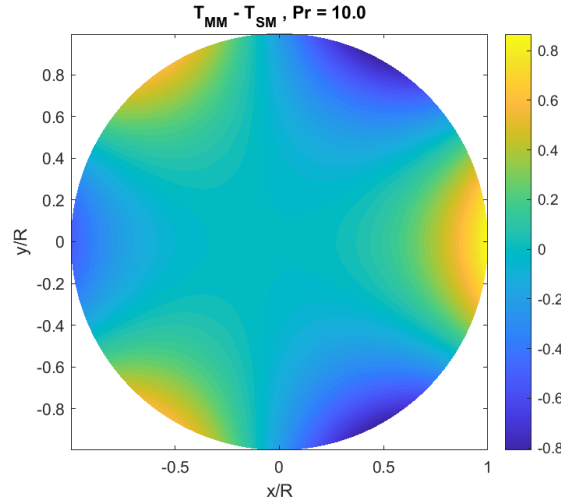


Figure 5: Difference between multi-mode and single mode temperature distributions

paring figures 3 and 4 reveals that the additions of the first 2 modes changes the temperature distribution significantly. Figure 5 shows the difference between the two cases. This rises the questions as to how many modes needs to be included to achieve a sufficiently precises description.

3.4 Convergence Criteria and Practical Truncation

In this section we consider a boundary conditions that depend only on θ that the Fourier coefficients are constants.

$$T_{BC}^* = T_o + \sum_{m=1}^{+\infty} A_m \cos(m\theta) \quad (44)$$

The resulting temperature profile in the fluid is therefore given by

$$T^* = \frac{\text{Pr}}{2} \left(1 - r^{*+} \right) + \sum_{m=1}^{+\infty} A_m r^{*m} \cos(m\theta). \quad (45)$$

Each Fourier mode k decays as r^{*k} as one moves toward the channel axis. At a given radial position $r^* = a$, the contribution of mode k is

$$T_k(a) = A_k a^k \cos(k\theta) \quad (46)$$

For $a < 1$ (inside the channel), higher modes decay rapidly. For example:

r^*	Mode 1	Mode 2	Mode 3	Mode 4
1.0	A_1	A_2	A_3	A_4
0.8	$0.80A_1$	$0.64A_2$	$0.51A_3$	$0.41A_4$
0.5	$0.50A_1$	$0.25A_2$	$0.13A_3$	$0.06A_4$
0.2	$0.20A_1$	$0.04A_2$	$0.01A_3$	$0.002A_4$

Table 3: Spatial decay factors r^{*k} showing how mode k vanishes toward the axis. By $r^* = 0.2$, modes $k \geq 3$ contribute less than 1%.

Key observation: Higher modes are relevant only near the wall ($r^* \approx 1$). Deep in the channel interior ($r^* < 0.5$), mode k contributes less than 0.5^k times its boundary value, making truncation at mode n accurate far from the wall.

3.4.1 Amplitude Distribution and Realistic Cooling Patterns

In practice, boundary conditions seldom have equal-amplitude modes. Real cooling patterns (e.g., uneven cooling, localized heat removal) have **decreasing amplitude** with mode number:

$$|A_k| \sim \frac{1}{k^\gamma} \quad (47)$$

where $\gamma > 0$ reflects the smoothness of the boundary condition. For example:

- $\gamma = 1$ (square wave or piecewise constant): $|A_k| \sim 1/k$
- $\gamma = 2$ (triangular or piecewise linear): $|A_k| \sim 1/k^2$
- $\gamma = 3+$ (smooth, analytic BC): $|A_k|$ decays faster than $1/k^3$

For typical cooling channels with smooth (non-sharp) temperature transitions, expect $\gamma \geq 1$.

3.4.2 Global Error Bound with Amplitude Decay

We define the truncation at mode N by

$$\Delta_N = \sum_{m=N+1}^{+\infty} A_m r^{*m} \cos(m\theta). \quad (48)$$

If the boundary condition is smooth enough that $|A_k| \leq C/k^\gamma$, then truncating at mode N gives a global error bounded by

$$\max_{r^*, \theta} |\Delta_N| \leq \sum_{k=N+1}^{\infty} \frac{C}{k^\gamma} \cdot r^{*k} \quad (49)$$

At the wall ($r^* = 1$):

$$|\Delta_N|_{\text{wall}} \leq C \sum_{k=N+1}^{\infty} \frac{1}{k^\gamma} \quad (50)$$

For $\gamma = 1$ (square wave): $\sum_{k=N+1}^{\infty} 1/k \approx \ln(N)$ — slow convergence For $\gamma = 2$ (piecewise linear): $\sum_{k=N+1}^{\infty} 1/k^2 \approx 1/N$ — moderate convergence For $\gamma \geq 2.5$ (analytic BC): converges exponentially in N

3.4.3 Practical Truncation Criteria

Define a relative error tolerance δ_{tol} (e.g., 1%, 0.1%, or 0.01%). A mode n can be dropped if its maximum contribution is below the tolerance:

$$\max_{\theta} |A_n r^{*n} \cos(n\theta)| < \delta_{\text{tol}} \cdot T_{\text{scale}} \quad (51)$$

where $T_{\text{scale}} = \max(|T_o|, \text{Pr}/2)$ is the temperature scale.

This simplifies to:

$$\boxed{|A_n| < \delta_{\text{tol}} \cdot T_{\text{scale}} \Rightarrow \text{mode } n \text{ is negligible}} \quad (52)$$

For a given tolerance, one can determine the **truncation order** N_{trunc} as the smallest N satisfying this criterion.

3.4.4 Numerical Example: Transformer Oil with Realistic Cooling

Consider a cooling pattern that is piecewise linear around the circumference:

$$T_{\text{BC}}(\theta) = T_o + \Delta T \max(0, \cos \theta) \quad (53)$$

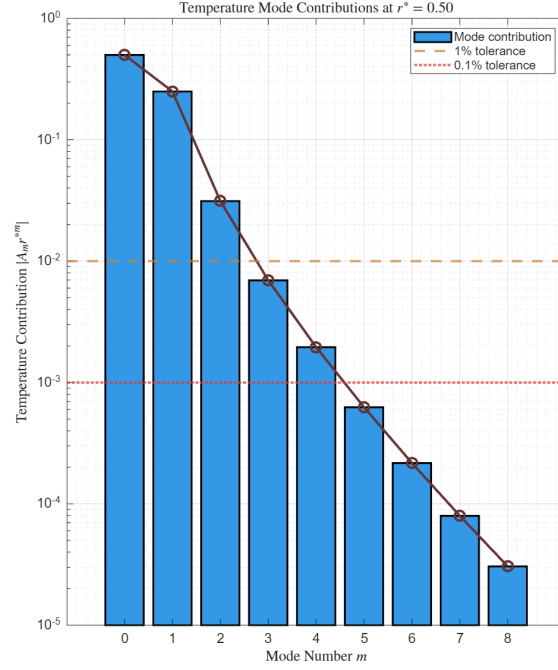
(i.e., cooling on one side, isothermal on the other).

This gives a Fourier series with $|A_k| \sim 1/k^2$. Suppose $\Delta T = 0.5$ (half the mean temperature) and $T_o = 1$. Then:

Mode n	1	2	3	4	5	6
A_n	0.25	0.025	0.008	0.004	0.002	0.001

Table 4: Fourier amplitudes and their max contributions assuming $\gamma = 2$ decay ($|A_n| \sim 1/n^2$).

Figure 6 shows the relative mode contributions for the case of piecewise linear cooling ($\gamma = 2$). Depending on the tolerance level we can deduce that for

Figure 6: Relative Mode contributions fro $\gamma = 2$

- for $\delta_{tol} = 0.1$, $N_{trun} = 2$.
- for $\delta_{tol} = 0.01$, $N_{trun} = 5$.

3.4.5 Decay Deep in the Channel

The spatial decay is crucial. At $r^* = 0.5$ (halfway to axis):

Mode n	1	2	3	4	5	6
Decay factor 0.5^n	0.50	0.25	0.125	0.06	0.03	0.016

Table 5: Spatial decay at $r^* = 0.5$. Mode contributions shrink by 2^n moving from wall to mid-channel. Mode 3 is suppressed by factor ~ 10 , mode 4 by ~ 16 .

With amplitude decay $|A_n| \sim 1/n^2$, at $r^* = 0.5$:

- Mode 1: $0.25 \times 0.50 = 0.125$ (large)
- Mode 2: $0.025 \times 0.25 = 0.006$ (small)
- Mode 3: $0.008 \times 0.125 = 0.001$ (negligible)
- Mode 4+: $\ll 0.001$

Physical implication: For calculating perturbative coupling effects on velocity (which depend on temperature gradients), modes beyond $n = 2$ are virtually irrelevant in the interior and only matter very close to the cooling boundary.

Result 3.1 (Convergence Summary). For laminar cooling channels with smooth boundary conditions:

1. **Mode 0 (axisymmetric):** Always dominant; captures uniform heating and viscous dissipation baseline. Contributes $\sim 50\%$ of overall temperature variation.
2. **Mode 1 (dipole):** Captures first-order azimuthal asymmetry (e.g., one-sided cooling). Contributes $\sim 30\text{--}40\%$ near wall; decays to $< 10\%$ at $r^* < 0.5$.
3. **Mode 2 (quadrupole):** Captures more complex patterns (e.g., alternating hot/cold spots). Contributes $\sim 5\text{--}20\%$ near wall if present; decays to $< 1\%$ by $r^* = 0.5$.
4. **Modes 3 and higher:** Typically $< 5\%$ contribution near wall; negligible ($< 0.1\%$) in interior. Needed only for highly non-smooth (sharp-cornered) cooling patterns or fine geometric details.

Practical recommendation for transformer cooling:

- For **1% accuracy**, truncate at $N = 2$ (retain modes $m = 0, 1, 2$)
- For **0.1% accuracy**, increase to $N = 4$ or $N = 5$
- For **$< 0.01\%$ accuracy**, $N = 6$ or $N = 7$

Since realistic cooling channels have smooth temperature profiles (not sharp discontinuities), $\gamma \geq 1.5$ is typical, making $N = 2$ sufficient for engineering purposes. This first analytical study of the relative importance of different Fourier modes, is essential for efficient simulation work. Knowing which Fourier modes are actually relevant to insure the tolerance level reduces computation time but eliminating negligible details.

4 Perturbative effects of temperature on flow profile

We now turn for the treatment of the central part of this work, That is the feedback of temperature variation on the momentum equations. To this end we start by making certain assumption that will simplify the analytical treatment.

4.1 Assumption & Definitions

The main assumption we make is that the temperature inhomogeneity in the fluid will affect only its viscosity. Generally one should also consider the dependence of density on temperature but given that we consider a horizontal channel and that gravity is not considered into this analysis we neglect the density dependence on temperature. Further more we will treat the problem perturbatively thus we

make the following definition

$$\mathbf{v}_t = \mathbf{v} + \delta\mathbf{v} \quad (54a)$$

$$P_t = P + \delta P \quad (54b)$$

$$\mu_t = \mu + \delta\mu \quad (54c)$$

The subscript 't' indicates the total quantities (sum of isothermal and perturbation). Further more we assume that the perturbation depend only on radial and azimuthal parameters r and θ . This because we consider boundary conditions that are independent of z . The only term that dependence directly on temperature is the viscosity variation $\delta\mu$, this can be derived from a given viscosity model as

$$\delta\mu = \left. \frac{d\mu}{dT} \right|_{T=T_o} (T - T_o), \quad (55a)$$

$$= a_\mu (T - T_o). \quad (55b)$$

Finally concerning the boundary conditions satisfied by the perturbations, we assume that the no-slip conditions are satisfied at the wall. Given the the velocity profile for the isothermal case already satisfies this conditions this leads to

$$\delta v_r(r = R), \quad \delta v_\theta(r = R) = 0, \quad \delta v_z(r = R) = 0. \quad (56)$$

4.2 The perturbation equations

The aim of this section is to derive the equations obeyed by the perturbations in dimensionless form.

Derivation 4.1. We start by recalling the definition of the total momentum flux tensor in covariant form

$$\Phi_t = \rho (\mathbf{v}_t \otimes \mathbf{v}_t) + P_t \mathbb{1} - \mu (\nabla \mathbf{v}_t + \nabla \mathbf{v}_t^\top), \quad (57)$$

where $\mathbb{1}$ is the identity operator. This can be then decomposed into isothermal momentum flux and perturbation flux by expanding the velocity, pressure and viscosity field.

$$\Phi_t = \Phi + \delta\Phi \quad (58)$$

The perturbation momentum flux tensor, $\delta\Phi$ is then given by

$$\delta\Phi = \rho (\mathbf{v} \otimes \delta\mathbf{v}) + \rho (\delta\mathbf{v} \otimes \mathbf{v}) + \delta P \mathbb{1} - \mu (\nabla \delta\mathbf{v} + (\nabla \delta\mathbf{v})^\top) - \delta\mu (\nabla \mathbf{v} + (\nabla \mathbf{v})^\top). \quad (59)$$

In general form the momentum conservation equations can be put into an invariant form through

$$\mathbf{o} = \nabla \Phi_t \quad (60a)$$

$$= \underbrace{\nabla \Phi}_{=\mathbf{o}} + \nabla \delta \Phi \quad (60b)$$

Thus the equation satisfied by the perturbations are

$$\nabla \delta \Phi = \mathbf{o} \quad (61)$$

The second equation is the of the mass continuity for incompressible flow satisfied by the perturbation. Expending equation 61 and incorporating the constraints $\nabla \cdot \delta \mathbf{v} = \mathbf{o}$, results in the tensor equation

$$\rho (\mathbf{v} \cdot \nabla) \delta \mathbf{v} + \rho (\delta \mathbf{v} \cdot \nabla) \mathbf{v} + \nabla \delta P - \mu \nabla^2 \delta \mathbf{v} - \nabla \delta \mu \cdot (\nabla \mathbf{v} + (\nabla \mathbf{v})^T) - \delta \mu \nabla^2 \mathbf{v} = \mathbf{o} \quad (62)$$

Recalling the expression of the gradient of a vector field

$$\nabla \delta \mathbf{v} = \begin{pmatrix} \partial_r \delta v_r & \frac{1}{r} (\partial_\theta \delta v_r - \delta v_\theta) & \partial_z \delta v_r \\ \partial_r \delta v_\theta & \frac{1}{r} (\partial_\theta \delta v_r + \delta v_r) & \partial_z \delta v_\theta \\ \partial_r \delta v_z & \frac{1}{r} \partial_\theta \delta v_z & \partial_z \delta v_z \end{pmatrix}, \quad (63)$$

and the Laplacien of a vector field in cylindrcal coordinates

$$\nabla^2 \delta \mathbf{v} = \begin{pmatrix} \nabla^2 \delta v_r - \frac{\delta v_r}{r^2} - \frac{2}{r^2} \partial_\theta \delta v_\theta \\ \nabla^2 \delta v_\theta - \frac{\delta v_\theta}{r^2} + \frac{2}{r^2} \partial_\theta \delta v_r \\ \nabla^2 \delta v_z \end{pmatrix}, \quad (64)$$

we can now recast equation 62 in its components we get

- **Radial Equation**

$$\partial_r \delta P - \mu \left(\nabla^2 \delta v_r - \frac{\delta v_r}{r^2} - \frac{2}{r^2} \partial_\theta \delta v_\theta \right) = \mathbf{o}. \quad (65)$$

- **Azimuthal Equation**

$$\frac{1}{r} \partial_\theta \delta P - \mu \left(\nabla^2 \delta v_\theta - \frac{\delta v_\theta}{r^2} + \frac{2}{r^2} \partial_\theta \delta v_r \right) = \mathbf{o}. \quad (66)$$

- **Axial Equation**

$$\rho \delta v_r \partial_r v_z - \mu \nabla^2 \delta v_z - (\partial_r \delta \mu) (\partial_r v_z) - \delta \mu \nabla^2 v_z = \mathbf{o} \quad (67)$$

In dimensionless form these equation are

- **Dimensionless Radial Equation**

$$\partial_{r^*} \delta P^* = 4\epsilon \left(\nabla_{\perp}^{*2} \delta v_r^* - \frac{\delta v_r^*}{r^{*2}} - \frac{2}{r^{*2}} \partial_{\theta} \delta v_{\theta}^* \right). \quad (68)$$

- **Dimensionless Azimuthal Equation**

$$\frac{1}{r^*} \partial_{\theta} \delta P^* = 4\epsilon \left(\nabla_{\perp}^{*2} \delta v_{\theta}^* - \frac{\delta v_{\theta}^*}{r^{*2}} + \frac{2}{r^{*2}} \partial_{\theta} \delta v_r^* \right). \quad (69)$$

- **Dimensionless Axial Equation**

$$\nabla_{\perp}^{*2} \delta v_z^* + 2r^* \text{Re} \delta v_r^* = 2r^* \partial_r^* \delta \mu^* + 4\delta \mu^*. \quad (70)$$

The transverse dimensionless Laplacian, $\nabla_{\perp}^{*2}$ is the operator

$$\nabla_{\perp}^{*2} = \frac{1}{r^*} \partial_{r^*} (r^* \partial_{r^*}) + \frac{1}{r^{*2}} \partial_{\theta}^2 \quad (71)$$

Physical Remark. In the previous derivation note that the the temperature effects (through the perturbation of viscosity) only act as a source term for the axial momentum conservation equation. In fact the radial⁶⁸ and azimuthal ⁶⁹ equation along with the continuity equation form a closed system of equation that can be solved independently of the axial equation ⁷⁰. We should note that this in general not the case but rather a consequence of the assumption made on the velocity perturbation independence of the axial variables. Thus the strategy to solve this systems of equation will be to solve the closed subsystem of the radial azimuthal and continuity equation then inject the result for the radial perturbation in the axial equation.

4.3 Solving the Perturbation Equations

Derivation 4.2. To solve the equations derived in the previous section we first star by combining the radial and azimuthal equation into one cross sectional vector equation by defining the cross sectional divergence operator

$$\nabla_{\perp}^* = \begin{pmatrix} \partial_{r^*} \\ \frac{1}{r^*} \partial_{\theta} \end{pmatrix}, \quad (72)$$

and the cross sectional velocity field

$$\delta \mathbf{v}_{\perp}^* = \begin{pmatrix} \delta v_r^* \\ \delta v_{\theta}^* \end{pmatrix} \quad (73)$$

Then we can summarize equations 68 and 69 into

$$\nabla_{\perp}^* \delta P^* = 4\epsilon \nabla_{\perp}^{*2} \delta \mathbf{v}_{\perp}^* \quad (74)$$

Let ω be the axial vorticity defined by

$$\omega^* = \nabla_{\perp}^* \times \delta \mathbf{v}_{\perp}^*. \quad (75)$$

Note that ω is a scalar.

Taking the cross sectional curl of equation 74 we get the equation

$$\nabla_{\perp}^{*2} \omega^* = 0 \quad (76)$$

Finally we define the stream function ψ as

$$\delta \mathbf{v}_{\perp}^* = \begin{pmatrix} \frac{1}{r^*} \partial_{\theta} \psi \\ -\partial_{r^*} \psi \end{pmatrix}. \quad (77)$$

Then using the definition of vorticity 75

$$\omega^* = \nabla_{\perp}^* \times \delta \mathbf{v}_{\perp}^*, \quad (78a)$$

$$= \nabla_{\perp}^{*2} \psi. \quad (78b)$$

Hence the stream function ψ satisfies the biharmonic equation

$$\nabla_{\perp}^{*4} \psi = 0. \quad (79)$$

The general solution for this equation that satisfies regularity constraints at the origin is

$$\psi = \sum_{m \geq 1} \left(a_m r^m + c_m r^{*m+2} \right) e^{im\theta}. \quad (80)$$

Injecting this into equation 77 we get

$$\delta v_r^* = i \sum_{m \geq 1} m \left(a_m r^{m-1} + c_m r^{*m+1} \right) e^{im\theta}, \quad (81a)$$

$$\delta v_{\theta}^* = - \sum_{m \geq 1} \left(m a_m r^{m-1} + (m+2) c_m r^{*m+1} \right) e^{im\theta} \quad (81b)$$

Then the boundary conditions at $r^* = 1$ lead to $\forall m \geq 1$

$$a_m + c_m = 0 \quad (82a)$$

$$m a_m + (m+2) c_m = 0 \quad (82b)$$

which lead to $a_m = c_m \forall m \geq 1$. **Thus, we deduce that the the no slip boundary condition**

impose that the transverse perturbation of the velocity field are null. Noting that $\delta v_r^* = 0$, the axial equation simplifies to

$$\nabla_{\perp}^{*2} \delta v_z^* = 2r^* \partial_{r^*} \delta \mu^* + 4\delta \mu^*. \quad (83)$$

Noting that $\delta \mu^* = \frac{a\mu}{\mu} \Delta T^*$ we get

$$\nabla_{\perp}^{*2} \delta v_z^* = 2 \frac{a\mu}{\mu} (r^* \partial_{r^*} \Delta T^* + 2\Delta T^*) \quad (84)$$

Consider a general case for which the temperature profile is given by

$$\Delta T^* = \frac{\text{Pr}}{2} (1 - r^{*4}) + \sum_{m=1}^{\infty} A_m r^{*m} \cos(\theta) \quad (85)$$

Hence the right hand side of equation 84 becomes

$$\nabla_{\perp}^{*2} \delta v_z^* = 2 \frac{a\mu}{\mu} \left[\text{Pr} (1 - 3r^{*4}) + \sum_{m=1}^{\infty} (m+2) A_m r^{*m} \cos(m\theta) \right] \quad (86)$$

We will look for solutions of the form

$$\delta v_z^* = u_0(r) + \sum_{m=1}^{+\infty} u_m(r) \cos(m\theta). \quad (87)$$

Equation 84 then results in the following system of equation

$$\frac{1}{r^*} \partial_{r^*} (r^* \partial_{r^*} u_0) = 2 \frac{a\mu}{\mu} \text{Pr} (1 - 3r^{*4}) \quad (88a)$$

$$\frac{1}{r^*} \partial_{r^*} (r^* \partial_{r^*} u_1) - \frac{m^2}{r^{*2}} u_1 = 2 \frac{a\mu}{\mu} A_m (m+2) r^{*m} \quad (88b)$$

The boundary conditions and regularity requirements result in the following conditions

$$u_0(r^* = 1) = u_1(r^* = 1) = 0, \quad u_0(r^* = 0) \text{ and } u_1(r^* = 0) \text{ are finite}$$

Thus, the solutions of this system of equations is

$$u_0(r^*) = 2 \frac{a\mu}{\mu} \text{Pr} \left(\frac{r^{*2}}{4} - \frac{r^{*6}}{12} - \frac{1}{6} \right), \quad (89a)$$

$$u_m(r) = 2 \frac{a\mu}{\mu} A_m \frac{m+2}{4(m+1)} \left(r^{*(m+2)} - r^{*m} \right) \quad (89b)$$

4.4 Solution Analysis & conclusions

Before moving with an analysis of the effects of viscosity perturbation on the flow profile, we first take a step back to delve deeper into the conclusions of the previous section. The main result is that

temperature induced viscosity variation do not result in a recirculation of the flow in the radial and azimuthal directions. This is however a direct result of the initial assumptions regarding the velocity profile, the assume independence of axial parameter ie $\partial_z \delta \mathbf{v} = 0$. This is however not always the case and this assumptions needs to be revisited for more realistic flows. The second point is regarding the validity of the perturbative analysis, up to this stage we haven't specified in what conditions is the perturbative analysis valid. The requirement is that the perturbation is much smaller that the isothermal value, for viscosity this is summarized by

$$\delta \mu^* \ll 1 \quad (90a)$$

$$\left| \frac{a_\mu}{\mu} \Delta T_{max}^* \right| \ll 1 \quad (90b)$$

For the case of one sided cooling, dicussed in the previous section this equivalent to

$$\left| \frac{a_\mu}{\mu} \left(\frac{\text{Pr}}{2} + A \right) \right| \ll 1 \quad (91)$$

Thus for small Pr numbers and mode amplitude this condition is satisfied and the perturbative analysis is valid. Finally regarding the parameter $a_\mu = \frac{d\mu_t}{dT}$, we note that this parameter is negative for fluids, reflect the fact that viscosity decreases with temperature.

4.4.1 Example: One sided cooling

We propose to compare the isothermal flow with the flow with one sided cooling for which the temperature profile is the following

$$\Delta T^* = \frac{\text{Pr}}{2} (1 - r^{*4}) + A r^* \cos(\theta) \quad (92)$$

Thus using the general solution derived in the previous section, we get

$$u_0(r^*) = 2 \frac{a_\mu}{\mu} \text{Pr} \left(\frac{r^{*2}}{4} - \frac{r^{*6}}{12} - \frac{1}{6} \right), \quad (93a)$$

$$u_1(r^*) = 2 \frac{a_\mu}{\mu} \frac{3}{8} A (r^{*3} - r^*). \quad (93b)$$

Figure 7 shows the perturbation field and relative perturbation strength compared to the isothermal flow. Note that the perturbative velocity field correspond to around 3% acceleration in center of the channel this is because the temperature is highest in the center and thus the viscosity is the lowest hence the shear force is the weakest. Figure 8 shows the radial functions u_0 and u_1 , we see that the most important correction comes from the 0 order mode.

5 Conclusion

This work develop a perturbative analysis of the temperature effects on laminar flow in a channel with aim for application in transformer oil. We started by solving the steady state isothermal velocity field,

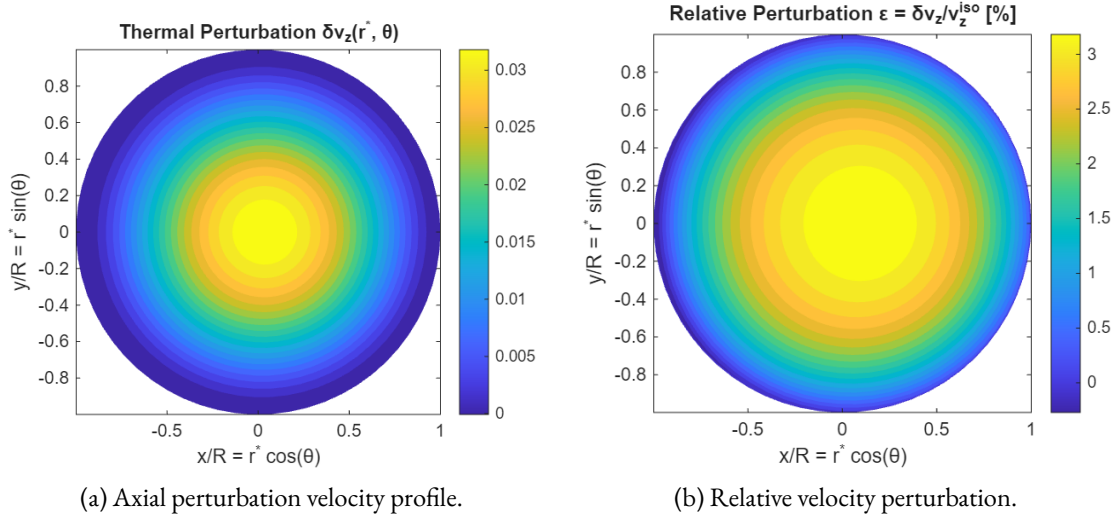


Figure 7: Effect of non-uniform cooling on the axial velocity profile. The left panel shows the perturbation velocity field, while the right panel shows the relative importance of the perturbation to the isothermal case. $Pr = 10$, $\frac{\alpha\mu}{\mu} = -0.01$ and $A = 0.5$.

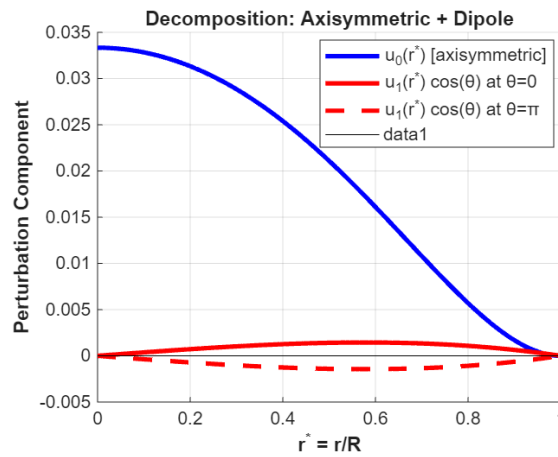


Figure 8: Axial and dipole perturbation.

then injecting this into the energy equation which allowed for a derivation of the temperature profile inside the channel for a different range of non uniform boundary conditions. We inspect the relative effects of different modes in the Fourier expansion of the boundary conditions. And concluded that for most engineering application considering the first 5 modes provide sufficient precision because modes decay away from the wall. The final step was coupling the temperature field back into the momentum equation through viscosity (assume density states invariant). We assumed that is back reaction results in a perturbation to the velocity and pressure fields in the channels. We showed that if we assume that the perturbation does vary axially then the radial and azimuthal perturbation vanish and the temperature profile only results in perturbation to the axial velocity. Finally this perturbation is solved for. We showed that the perturbation is dominated by the axial mode.

6 Future Work

We aim at expanding this analysis by first comparing our results to CFD simulations to see where exactly does perturbative linear analysis stop being efficient. We also plan at dropping the assumption regarding the independence of the perturbation field of the axial parameter. An other interesting point will be to couple this analysis to charge transport and see how temperature gradient effect the isothermal charge transport.

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A Cylindrical Laplacian

The full Laplacian in cylindrical coordinates is:

$$\nabla^2 T = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2}.$$

The transverse (radial + angular) Laplacian is:

$$\nabla_{\perp}^2 T = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2}.$$